A-A-50492
28 February 1989

COMMERCIAL ITEM DESCRIPTION

OILERS, AIR LINE

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for Military Specification MIL-O-82042 which is canceled.

Abstract. This CID covers two types of air line oilers as follows:

Type I - Combination oiler with filter, pressure regulator, and gauge. Size 1/2 - 1/2-inch (12.7 millimeters (mm)), internal, pipe thread connections.

Type II - Combination oiler and filter. Size 1/4 - 1/4-inch (6.35 mm), internal, pipe thread connections.

Salient characteristics.

1. Type I. The type I combination oiler shall consist of a filter, pressure regulator, gauge, and an oiler (lubricator) similar to figure 1. The type I oiler shall be capable of removing at least 90 percent of the condensed water and 95 percent of 50 micron or larger size foreign particles from an air line delivering up to 70 standard cubic feet per minute (scfm) (1.98 cubic meters per minute (m^3/m)) at an inlet pressure of 100 pounds per square gauge (psig) (689.5 kilopascals (kPa)). The unit shall also regulate the air pressure between 10 and 125 psig (68.9 and 861.9 kPa) and dispense a constant, adequate supply of lubricant to the discharge air in the air supply system. Oilers may have metal or plastic bowls with bowl guards. Oilers with metal bowls shall be capable of withstanding pressures up to 250 pounds per square inch gauge (psig) (1,723.7 kPa) at a temperature of 100° Fahrenheit (F) (37.7° Centigrade (C)).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4930

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Oilers with plastic bowls shall be capable of withstanding pressures up to 150 psig (1,034.2 kPa) at a temperature of 100°F (37.7°C).

The oiler shall have a body, bowl, sight dome or sight glass, adjustable oil feed, filler plug, gaskets, and other necessary components. The sight glass shall be constructed of a readily removable, shatter-resistant, plastic or pyrex material. A filler plug, provided with means for bleeding off pressure inside the oil chamber before removal of the plug, shall be readily accessible for refilling. A metering valve shall be furnished to accomplish adjustment of the oil feed rate from a flow of one drop per minute to maximum flow. The valve shall not change its setting as a result of jars or vibrations, such as those normally encountered in air supply systems. The oiler shall function under continuous or intermittent operation and shall not cause flooding of oil into the airline. The amount of atomized lubricant injected into the air shall be proportional to the amount of air flowing through the system. Oilers shall be capable of operating with oil having a viscosity up to 800 Saybolt Universal Seconds equivalent to SAE No. 30 oil at a temperature not less than 85°F (29.4°C). Type I oiler shall have a capacity of not less than 8 ounces (236.6 mm). Type I oiler shall have a maximum pressure drop of 6 pounds per square inch (psi) (41.4 kPa) at an inlet pressure of 100 psig (689.5 kPa) and 70 scfm (1.98 m^3/m). When specified, the oiler shall be so constructed that refilling of the reservoir can be accomplished without shutting off the air supply.

The filter shall be constructed with a body, transparent bowl, filter element, drain, gaskets, and other necessary components. The bowl shall be readily removable without the use of tools and constructed of shatter-resistant plastic, such as polycarbonate resin. A drain shall be provided to allow liquids and foreign matter to be drained from the bowl. Unless otherwise specified, the drain shall be of the manual type. When specified, the drain shall be of the automatic type. The drain shall be so constructed that it cannot be disassembled when opened for drainage purposes. The design of the filter shall be such that liquids and foreign matter (dirt, rust, scale) will be separated from the compressed air and dropped to the bottom of the bowl. Means shall be provided to prevent accumulated water and foreign matter from re-entering the air stream and making contact with the filter element. Type I filters shall have 1/2-inch (12.7 mm) internal inlet and outlet connections, a bowl capacity of at least 8 ounces (236.6 mm), and a maximum pressure drop of 3 psi (20.7 kPa) at 100 psig (689.5 kPa) inlet pressure and 70 scfm (1.98 m^3/m). Flows and pressure drops specified shall be with new clean elements. Unless otherwise specified, the filter elements shall be porous plastic or porous bronze or impregnated felt or other permanent and cleanable material. Unless otherwise specified, the filter elements shall have a nominal rating of 50 microns or less. The filter element shall have means to prevent bypass of contaminants and shall be readily removable without use of tools.

The pressure regulator shall have a body, a hand-operated adjusting device. The body shall be equipped with two 1/2-inch (12.7 mm) internal thread ports, and two 1/4-inch (6.35 mm) internal thread gauge ports furnished with removable plugs. The pressure adjustment shall be such that an inlet

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pressure of 250 psig (1,723.7 kPa) can be reduced and regulated within the range of 10 psig to 125 psig (68.9 kPa to 861.9 kPa). Internal components shall be constructed of corrosion-resistant materials. Unless otherwise specified, regulators shall be of a relieving type. While the regulator is subjected to 250 psig (1,723.7 kPa) constant primary air pressure, and with the free flow rates ranging from 0 to 70 scfm (0 to 1.98 m^3/m), the regulated pressure shall vary not more than 10 psig (68.9 kPa) at 100 psig (689.5 kPa).

The pressure gauge shall have a 2-inch (50.8 mm) or larger size dial having a minimum pressure range of 0 to 160 psig (0 TO 1,103.2 kPa) and a back connection fitted with a 1/4-inch (6.35 mm) external thread. The gauge shall have a case with a commercial finish and a shatter-resistant plastic crystal. The dial shall be numbered at intervals indicting 20 psig (137.9 kPa) pressure and marked at 5 psig (34.5 kPa) primary divisions. Numerical measurements shall be accurate within 2 percent of the full scale value in the middle half section of the scale and 3 percent of the full scale value in the first and last of the four sections. When subjected to a pressure of not less than 1.3 times the maximum scale reading for 1/2 hour, the gauge shall withstand the pressure without damage or change in accuracy. The gauge shall retain a minimum pressure of 80 percent of the full scale reading for a period of 1/2 hour with no loss in the indicated pressure.

The oilers shall be completely assembled and ready for operation. The equipment shall be treated and painted in accordance with the manufacturer's standard practice. The manufacturer's standard marking shall be on all units in accordance with the manufacturer's standard practice.

When specified and when applicable, the following accessories shall be furnished:

a. Metal or plastic bowl guard for filter and oiler.
b. Metal bowl suitable for pressures of 250 psig (1,723.7 kPa) and temperatures of 150°F (65.6°C).
c. Mounting bracket for regulator to be used with type I.
d. Mounting bracket for filter to be used with type II.

2. Type II. The type II oiler shall consist of a filter and oiler. The type II oiler shall conform to all of the requirements for type I oiler except for the following requirements.

The type II oiler shall be capable of removing at least 90 percent of the condensed water and 95 percent of 50 micron or larger size foreign particles from an air line delivering up to 20 scfm (.57 m^3/m) at an inlet pressure of 100 psig (689.5 kPa). The unit shall function in an air supply between 10 psig and 125 psig (68.9 and 861.9 kPa) and dispense a constant adequate supply of lubricant to the discharge air in the air supply system. The type II oiler shall have an oil capacity of not less than 4 ounces (118.3 mm). The type II oiler shall have a maximum pressure drop of

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6 pounds per square inch (psi) (41.4 kPa) at an inlet pressure of 100 psig (689.5 kPa) and 20 scfm (.57 m^3/m). Type II filters shall be 1/4-inch (6.35 mm) internal connections, a bowl capacity of at least 4 ounces (118.3 mm), and a maximum pressure drop of 3 psi (20.7 kPa) at 100 psig (689.5 kPa) inlet pressure and 20 scfm(.57 m^3/m).

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

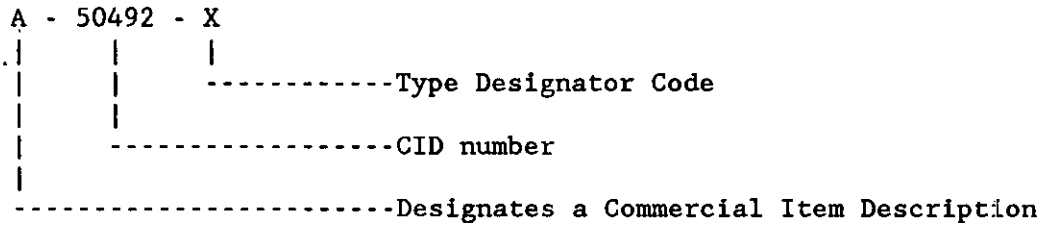
Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

CID based part identification numbers. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



The type of airline oiler is identified by a single numerical code character, see table I.

TYPE	Type Designator Code
I	1
II	2

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Notes.

Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document. b. Type of oiler required.
c. Oiler capable of being refilled without shutting off air supply, when required.
d. When automatic drain is specified.
e. Filter element material and nominal filtration rating in microns, if different.
f. Pressure regulator, if different.
g. Required accessories.

MILITARY INTERESTS:

Custodians

Navy - YD

Review Activities

DLA - CS

CIVIL AGENCY COORDINATING ACTIVITIES

GSA - FSS

PREPARING ACTIVITY:

Navy - YD

Project Number 4930-0330

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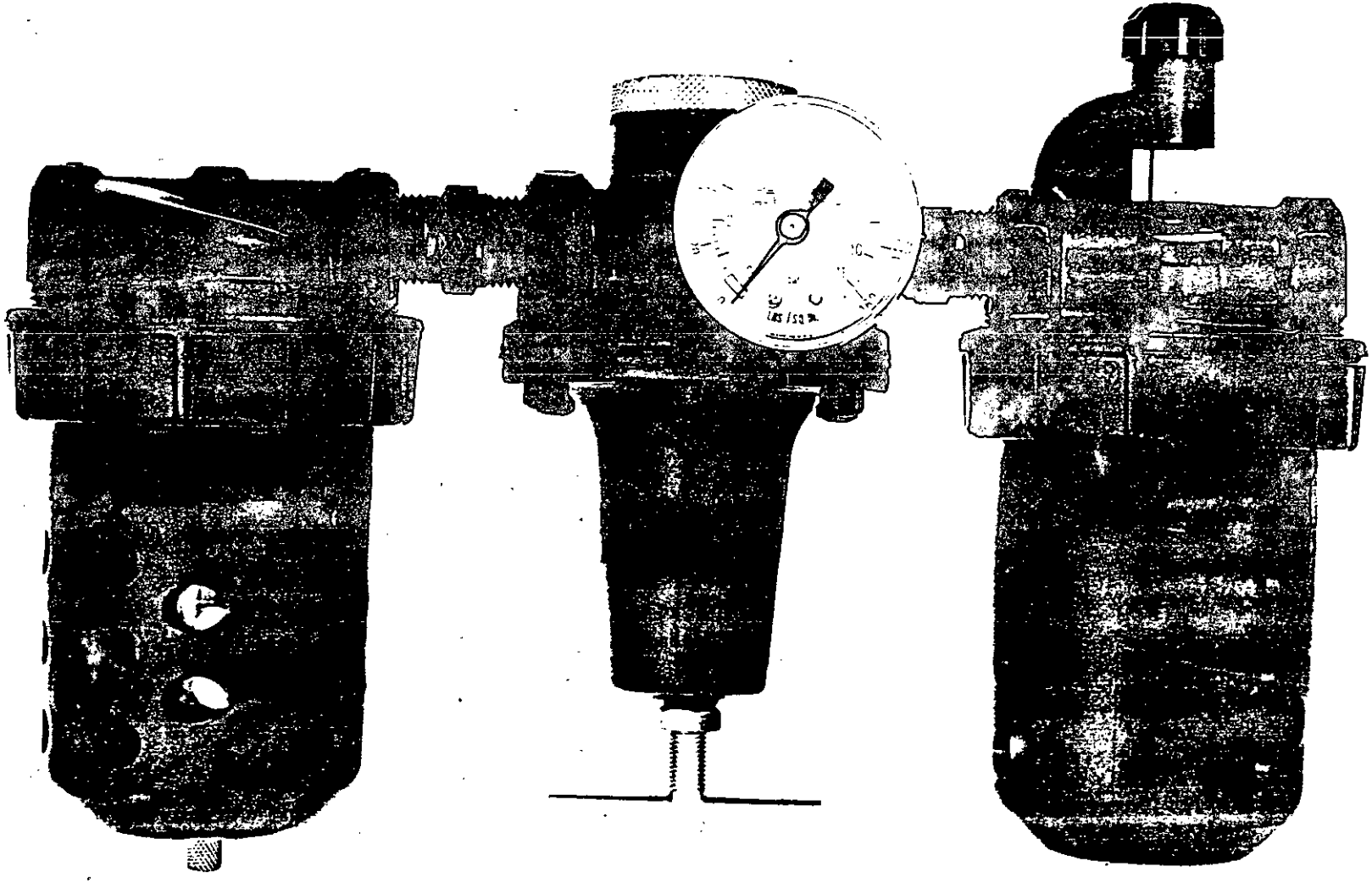


FIGURE #1 - TYPE I, OILER, FILTER, REGULATOR, GAUGE

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